

FOX_Checklist

What this is: every FoxByte feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `FOX_Feature_Implemented.pdf` (same ids).

Last updated: 21 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection (`chore/foxbyte`)

Source: `docs/FOX_Checklist.html` → `make feature-doc` → this PDF

☒ Done & tested — automated tests (unit, VM integration or CI)

☒ Done — tested by hand only

☐ Partial — gaps, or not verified end to end

☒ Known issue

☐ Not started

Sub-features: ☒ built ☐ partly ☐ not built yet ☒ broken · "VM §n" = section of `make integration` ; "v2 §n" = `make integration-v2` ; "update" = `make integration-update`

Where we are

Area	Features	Done & tested	Done (manual)	Partial	Known issue	Not started	Progress
A · Install & setup	9	4	3	2	.	.	78%
B · CLI & services	2	2	.	.	.	.	100%
C · Branching & storage	8	7	.	1	.	.	88%
D · Durability	6	5	.	1	.	.	83%
E · Security foundations	3	3	.	.	.	.	100%
F · SQL gateway	7	6	1	.	.	.	100%
G · Accounts & auth	5	4	.	1	.	.	80%
H · Blackbox (record of schema changes)	18	17	.	1	.	.	94%
I · Import & migration	13	5	8	.	.	.	100%
J · ETL pipelines (experimental)	4	2	2	.	.	.	100%
K · APIs, SDKs & agents	6	4	2	.	.	.	100%
L · Web console	11	1	10	.	.	.	100%
M · Build, CI & releases	7	6	1	.	.	.	100%
Total (features built or started)	99	66	27	6	.	.	94%
N · Not started (roadmap)	11	.	.	.	.	11	0%

Storage engine — not implemented, by design

FoxByte does **not** have its own storage engine. Data is stored by **PostgreSQL's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:18-bookworm` image plus `wal-g` (`postgres:16-bookworm` for an install created before 21 Sep 2026). This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.

What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with `wal-g` + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

Suggested next steps

- Run on real machines what the throwaway VM cannot: `fox update` across a real release on macOS and Windows (A9); on Windows, the `zstd` distro image's unpack (M5), the btrfs holding area of `fox pg upgrade` (D6), and an export's copy to the host (D5).
- Make the `postgres-wal-g` image package public, so installs pull it instead of building it for a few minutes (C8).
- Run the VM integration suites in CI on every pull request (M2).

4. Add automated tests where there are none: imports (I1–I3, I7), SDKs (K3, K4), web UI (L*).
5. HA beyond the demo: automatic failover and fencing (D4).

A Install & setup

7 of 9 done

ID	Feature · sub-features	Tested by
 A1	One-line installer (macOS / Linux) ✓ Detects OS / CPU, downloads the release binary ✓ Installs to the prefix (sudo only if needed) ✓ Stages the Linux engine for the VM on macOS	Unit (<code>test_install_sh.sh</code> , CI); manual
 A2	Windows installer (install.ps1) ✓ Installs WSL, elevates, resumes after reboot ✓ Downloads fox.exe, engine, image build context ✓ Ubuntu rootfs verified against Ubuntu's checksums ✓ Adds to PATH and runs <code>fox setup</code>	Pester (Windows CI); Windows run 16 Sep
 A3	fox setup on macOS (Lima) ✓ Creates / starts the Lima VM ✓ Installs Docker and ZFS ✓ Reinstalls the engine on every run ✓ Starts the stack	Unit; run against a deliberately broken VM, 21 Sep
 A4	fox setup on Windows (WSL2 + btrfs) ✓ Imports a dedicated distro with systemd ✓ Docker, btrfs and a storage mount unit ✓ Quiet steps, detail in install.log ✓ Waits for systemd on a cold distro	Unit (<code>host_wsl_test.go</code>); Windows run 16 Sep
 A5	Linux native ✓ Engine runs in-process, no VM	VM integration (all suites run on Linux)
 A6	Command forwarding & host file streaming ✓ macOS → Lima VM, starts the VM if stopped ✓ Windows → WSL distro	Unit (<code>TestImportLocalFile</code> , <code>TestWSLArgs</code>)
 A7	Always-latest engine binary ✓ Downloads the newest engine into the cache (ELF check) ✓ <code>FOX_NO_REFRESH</code> , <code>FOX_VERSION</code>	Unit (<code>host_test.go</code> , <code>update_test.go</code>); Windows run 16 Sep
 A8	New-release notice on fox start ✓ Background check, 1.5 s limit, printed last ✓ Only a promoted, newer, complete release	Unit; update \$1, \$1b, \$2, \$3, \$8; real Mac check
 A9	fox update ✓ Downloads and verifies everything before changing anything ✓ Stops servers, installs engine, restarts old one on failure ✓ <code>up</code> , Blackbox upgrade, <code>start</code> on the new engine ✓ Replaces host <code>fox</code> , keeps previous binaries	Unit; update \$4–\$7; a real Mac run

B CLI & services

2 of 2 done

ID	Feature · sub-features	Tested by
----	------------------------	-----------

✓	B1 The fox command set	✓ Stack: setup, start, stop, up, down, status, logs, psql, update ✓ Branching, backup / restore, HA ✓ Blackbox, policy, impact ✓ Import, pipelines ✓ backup export / restore, pg status / upgrade	✓ Users, API keys, admin ✓ <code>fox vm status / fox vm shell</code> (Lima, WSL; Linux says there is no VM) ✓ <code>branch create --from</code> ✓ <code>fox uninstall</code> removes it all, retired installs included	Unit (<code>main_test.go</code> , <code>update_test.go</code>); VM suites
✓	B2 Background services (daemon)	✓ Control plane :8080, gateway :6432, agent API :8088 ✓ Detached, pidfiles and logs in <code>~/ .fox</code>	✓ Stop (TERM then KILL), status, logs □ Health check / restart on crash	Unit (<code>TestStopTerminatesProcess</code>); VM §1

C Branching & storage

7 of 8 done

ID	Feature · sub-features	Tested by
✓ C1	Storage drivers (ZFS / btrfs) ✓ ZFS snapshot + clone per branch (macOS VM, Linux) ✓ btrfs subvolume snapshots (Windows) ✓ Pool / filesystem created on a loopback file	Unit; VM §2, §2b (ZFS); btrfs manual on Windows
✓ C2	Branch lifecycle ✓ Create, list, delete ✓ Reset to parent ✓ Suspend / resume ✓ Recreate a container from its surviving data	Unit; VM §2, §2b, §4
✓ C3	Protection of main ✓ Delete and reset refused	Unit; VM §4, §6
✓ C4	Private networking (no host ports) ✓ Branch databases reachable only through the gateway	Unit (<code>TestParsePublishedPort</code> , <code>TestDSN</code>)
✓ C5	Scale-to-zero & wake on connect ✓ Idle branches suspended automatically ✓ Next connection wakes the branch (~0.3–0.7 s)	VM §7
🔗 C6	Storage quotas ✓ Space reserved for main (ZFS) 🔗 Per-branch cap (opt-in)	Manual
✓ C7	Capacity & status ✓ Services, main health, backups, branches ✓ Storage used / available	VM §1
✓ C8	Provisioning & image pull ✓ Docker, storage pool and image prepared on <code>up</code>	Unit; VM §11b

D Durability

5 of 6 done

ID	Feature · sub-features	Tested by
✓ D1	WAL archiving to object storage ✓ wal-g ships main's WAL to MinIO, off the commit path	VM §3
✓ D2	Base backups ✓ Create and list ✓ Follow the primary after a failover; listing works with main stopped	VM §3, §3b, §6

✓	D3 Point-in-time restore	✓ Restore to a timestamp or latest on :5433 ✓ Per-install credentials ✓ Starts from the newest base backup that precedes the target	✓ A point no backup precedes is refused, naming the oldest	Unit; VM §3, §3b, §6; v2 §1a
🔥	D4 HA standby & failover	✓ Streaming hot standby ✓ Manual failover, <code>main</code> rerouted ✓ Nothing deletes or stops the promoted standby ✓ <code>fox up</code> brings it back, not the old main	✓ <code>fox ha failback</code>, keeping every write ✓ The promoted standby archives WAL, so backups and PITR cover the failover ❑ Automatic failover, fencing, multi-host	Unit; VM §6, §7
✓	D5 Portable export & restore	✓ Every non-agent branch: <code>pg_dump</code>, its roles, the Blackbox head; SHA256SUMS over all ✓ Copied out of the VM and verified on the host; a copy that fails is deleted ✓ Damaged, incomplete or path-escaping archives refused	✓ Restore into a new branch, same or newer major; Blackbox checked exact 🔥 Host copy on macOS/Windows unit-tested only	Unit; VM pg-upgrade §1
✓	D6 PostgreSQL major upgrade	✓ <code>fox pg status</code>: the major of the install and of each branch ✓ Plan with every change crossed, found-in-your-data first; refusals say what to do ✓ Asks, exports, then reloads every carried branch on the new major	✓ Data, roles, API keys and every Blackbox entry come across ✓ Old databases held outside the branch namespace; rollback and finalize ✓ Each upgraded cluster archives WAL under its own prefix 🔥 <code>btrfs</code> (Windows) holding area not yet run ❑ Carry agent branches; upgrade with HA on	Unit; VM pg-upgrade (72 / 72)

E Security foundations 3 of 3 done

ID	Feature · sub-features	Tested by
✓	E1 Per-install generated secrets ✓ Generated Postgres and MinIO passwords	Unit (indirect); v2 §1a
✓	E2 TLS everywhere ✓ Self-signed TLS on all three services, zero setup	Unit (<code>TestReadStartupUpgradesTLS</code>)
✓	E3 Keys are made by a person, not by start ✓ Start creates no account and no key, and caches none on disk ✓ The banner shows the connection string and how to make a key	Integration §1
✓		

F SQL gateway 7 of 7 done

ID	Feature · sub-features	Tested by
✓	F1 Routing by database name ✓ One endpoint :6432, database name = branch	Unit; VM §1, §6
✓	F2 API-key authentication ✓ <code>key_</code> key as the Postgres password	VM §1
✓	F3 Backend login (SCRAM) ✓ Cleartext, md5, SCRAM-SHA-256 ✓ Clients never see the database password	VM §1, §6, §8 (exercised)
✓	F4 Per-user Postgres login roles ✓ Role per user, actor can't be forged	Manual
✓	F5 Attribution session settings ✓ Actor kind, branch, session id per connection	VM §8
✓		

✓	F6 Ctrl-C (CancelRequest) forwarding ✓ Cancel a running query through the gateway	Unit
✓	F7 Errors & insecure build tag ✓ Real Postgres error responses ✓ No-auth bypass impossible in release builds	Unit (TestSendError)

G Accounts & auth 4 of 5 done

ID	Feature · sub-features	Tested by
 G1	Auth store (SQLite) ✓ Users, keys, sessions, OAuth ids, pipelines ✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once □ Roles / permissions, multi-tenant projects	Unit (<code>TestVerifyKeyConcurrent</code> , <code>TestConcurrentFirstOpen</code>); VM §1
 G2	Email & password users ✓ Create account, log in □ Password change / reset □ List and delete users	Unit; VM §1
 G3	API keys ✓ Create (shown once), list, revoke ✓ Bearer / X-API-Key and gateway password  Revoking an unknown id reports success	Unit; VM §1
 G4	Web sessions & signup control ✓ httpOnly session cookie ✓ Signup can be closed	Unit (<code>TestSessions</code>)
 G5	OAuth (GitHub / Google) ✓ Sign-in flows built ✓ Defaults match what the engine serves (<code>https://localhost:8080</code> ; web origin = public URL) ✓ Session cookie <code>Secure</code> / <code>SameSite</code> follow the origin □ Tested with a real OAuth app	Unit; VM §1b

H Blackbox (record of schema changes) 17 of 18 done

ID	Feature · sub-features	Tested by
 H1	DDL recording ✓ Who, tool, branch, statement, status, risk ✓ Inherited by every branch ✓ A restart records nothing: the install script's own statements are not captured ❑ Role / database changes, some dropped object types	VM §8; v2 §9
 H2	Hash chain, append-only & verify ✓ Rows can't be edited or deleted normally ✓ fox ledger verify  Superuser rewrite caught only via anchors (H10–H12) ✓ bb.policy changes recorded (append-only, not chained)	VM §8; v2 §0, §6b
 H3	Attribution ✓ Humans via their login role ✓ Agents via their own agent-<id> role ✓ Kind derived from the role, so neither can pose as the other ✓ Tool and session are client-declared by design, and labelled so	Unit; VM §8; v2 §1c
 H4	Destructive-DDL guardrail ✓ DROP TABLE / DROP SCHEMA refused, attempt recorded ✓ Override only superusers / db_admin , use recorded ✓ "flag" and "allow" actions; unknown actions refused ✓ No hang when blocked after a write in the same transaction	VM §8; v2 §1b, §6b
 H5	Risk flags ✓ Type change, drop column flagged ✓ Judged on the running statement (no comments or literals) ✓ Automated test	Unit; v2 §6b
 H6	Blackbox access (CLI / REST / MCP / UI / SDK) ✓ CLI, REST with filters, MCP, web page, SDKs ❑ Per-user access control	v2 §0, §5

✓	H7	fox ledger revert --to	Unit (help text); VM §3 (the restore)
✓		- Help and output say it's a point-in-time restore of main - Points to H13 for a branch from before a change	
✓	H8	Override permission (db_admin) & upgrade	v2 §1b, §6
✓		- fox admin grant revoke list - First user granted automatically - fox blackbox upgrade [--all] - Grant / revoke from the web (admins only)	
✓	H9	Blackbox 2.0 capture	Unit; v2 §2
✓		- Transaction id and WAL position per entry - Fail-safe, append-only - Task, tool-call hash, override use - Kill switch not settable by clients	
✓	H10	Checkpoints & anchors	Unit; v2 §3, §3b
✓		- Merkle root in the database and in read-only anchor files - Published anchor format - On a schedule and on demand - Cloud write-once anchors (S3 Object Lock)	
✓	H11	fox-verify (independent verifier)	Unit; v2 §3
✓		- Checks a live branch or an export - Release binaries for every platform	
✓	H12	Blackbox integrity & export	v2 §3
✓		- INTACT / TAMPERED via CLI, REST, MCP - JSON-lines export - Streaming for very large ledgers	
✓	H13	Branch from just before a Blackbox entry	Unit; v2 §4
✓		- Restore + replay up to the change's transaction - CLI, REST, MCP, web panel - By transaction id or timestamp - main untouched	
✓	H14	Blackbox names & aliases	Unit; v2 §5
✓		- fox blackbox, /blackbox routes, MCP and SDK names - Every original name still works	
✓	H15	Policy gate (warn / block)	Unit; v2 §6, §6b
✓		- BBX02 warn / BBX01 block with JSON detail - CLI, REST, MCP, web page; custom rules - Blocked attempts recorded - Patterns match the running statement - Per-rule db_admin override - TRUNCATE	
✓	H16	Agent provenance & execute_change	Unit; v2 §7
✓		- Session, task, parent session on every agent change (also through the Gateway DSN) - fox blackbox sessions, REST - MCP execute_change with policy preview - Provenance is what the agent says (not verified)	
✓	H17	Impact analysis	Unit; v2 §8
✓		- Dependents (views, foreign keys, indexes, depth) - CLI, REST, MCP, inside execute_change - Other branches that have the object - Functions, types, schemas not recognised from SQL - Policy verdict and risk score	
✓	H18	Blackbox diff between branches	Unit; v2 §8
✓		- Fork point, changes only on each side - CLI, REST, MCP - Objects changed on both - Merge (diff only today)	

I Import & migration

13 of 13 done

ID	Feature · sub-features	Tested by
✓ I1	From Postgres - Schema and data via pg_dump - Zero-table result reported as an error - Automated test	Manual
✓ I2	From MySQL ≤5.7 / MariaDB - pgloader with type mapping (multi-arch image) - Automated test	Manual

☒	I3	From MySQL 8.x ✓ Native mysqldump path (caching_sha2 auth) ✓ Types, NULLs, unicode, PK	❑ Other indexes, foreign keys, defaults ❑ Automated test	Manual
☒	I4	From MongoDB ✓ Collection → typed table, exact name	✓ Nested objects / arrays kept as jsonb	VM §9, §10
☒	I5	From a .sql dump ✓ Runs the dump into a new branch ✓ Fails listing every failed statement (instance kept)	✓ Same for Postgres imports (I1) ✓ Automated test	Unit; VM §11
☒	I6	From JSON / NDJSON ✓ Columns inferred from top-level keys	✓ An invalid document fails the import, naming its element or line	Unit; VM §10, §11
☒	I7	From CSV ✓ Table from the header, exact column names	❑ Typed columns (all text today)	Manual
☒	I8	Continuous replication & cutover ✓ Initial copy + streaming (logical replication) ✓ Cutover keeps data (CLI) ✓ Status and cutover from REST and the Import page	✓ A replicating branch is never auto-suspended ✓ Automated test (VM §5b) and driven in the browser	Unit; VM §5b
☒	I9	Guard handling during import ✓ Guardrail paused, recording stays on	✓ Re-armed even after a crash	Manual
☒	I10	Live progress (CLI & SSE) ✓ Live log and progress bar	🟡 psql subprocess output not streamed	Manual
☒	I11	Import REST endpoints ✓ From a connection string or an uploaded file	❑ Target name validation	Manual
☒	I12	Web Import page ✓ Database and file panels with live console	✓ No longer claims a SQLite loader that doesn't exist	Manual
☒	I13	Exact identifier fidelity ✓ Case kept (<code>userId</code> , <code>leaseAiChats</code>)	🟡 Field order of JSON sources not kept	Unit; VM §10

J ETL pipelines (experimental)

4 of 4 done

ID	Feature · sub-features	Tested by
☒	J1 Pipeline engine ✓ Extract → raw schema → SQL models → tests, in a fresh branch ✓ <code>ref()</code> / <code>source()</code> templating	Unit; VM §9
☒	J2 Data-quality tests ✓ <code>not_null</code>, <code>unique</code>, <code>accepted_values</code>, <code>row_count_min</code>, <code>custom</code>	Unit; VM §9
☒	J3 Stored pipelines & run history ✓ Saved per user, run over the API, history ❑ Read a run's log	Manual
☒	J4 Pipeline editor (web) ✓ Create, edit, run, results, history	Manual

K APIs, SDKs & agents

6 of 6 done

ID	Feature · sub-features	Tested by
☒	K1 Control-plane REST API ✓ Branches, query, Blackbox, policy, impact, import, pipelines, keys	Unit; VM §1; v2 §0–§8

✓	K2 OpenAPI spec ✓ Served, Blackbox 2.0 endpoints documented ✓ Matches the code (import fields, status codes, auth routes, /blackbox aliases) ✓ Drift check in CI	Unit (drift check)
✓	K3 Python SDK ✓ Branches, query, Blackbox □ Import, pipelines, keys, agents □ Tests □ Publish to PyPI	Manual
✓	K4 TypeScript SDK ✓ Same client for Node and browsers □ Tests □ Publish to npm	Manual
✓	K5 Agent Branch API ✓ A branch per agent, own non-superuser role ✓ Provenance (task, parent, session) ✓ DSN reachable from the host (gateway-routed, branch-scoped key) ✓ Cap, listing and TTL count suspended branches	Unit; VM §5; v2 §1c, §7
✓	K6 MCP server ✓ Branches, run_sql, Blackbox tools ✓ policy_check, execute_change, impact, diff ✓ Requires an API key and acts as that account; a scoped key is held to its branch ✓ Client config & 16-tool reference (docs/mcp.md)	Unit; v2 §1c–§8

L Web console 11 of 11 done

ID	Feature · sub-features	Tested by
✓ L1	Embedded single-origin UI ✓ Served by the engine at :8080 □ Automated UI tests (whole area)	Manual
✓ L2	Login & signup ✓ Log in / sign up tabs, OAuth buttons	Manual
✓ L3	Dashboard ✓ Live tiles ✓ Branch table: create, suspend, resume, delete, copy DSN	Manual
✓ L4	SQL console & browser ✓ Schema sidebar, rows, structure, indexes ✓ Values shown as Postgres writes them (uuid, interval, inet, bytea) ✓ SQL editor ✓ System tables hidden by default, with a show button ✓ Runs as the signed-in user; one-run override switch ✓ Per-rule override for policy blocks	Manual; API in v2 §6
✓ L5	Blackbox page ✓ Entries per branch with badges ✓ Table / risk / date filters, and "Clear filters" ✓ Verify button; a row expands to its statement and session	v2 §5b; browser
✓ L6	API keys page ✓ Create, copy, list, revoke ✓ A new key is shown inside its connection string (E3's next step)	Manual; browser review 21 Sep
✓ L7	Landing, Guide & Docs pages ✓ Landing, developer guide, reference ✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)	Type-check + build
✓ L8	Blackbox integrity page ✓ Verify, checkpoint, export, branch-before ✓ Used signed in; every result matched the CLI ○ Entry list needs a refresh to show a new change	Type-check + build; browser review 16 Sep
✓ L9	Policies page ✓ Rules, warn/block, custom rules, preview, evaluations ✓ Used signed in; matched the CLI, wrote only to the chosen branch ✓ Non-admin change refused with a readable message ✓ Who can override: list, grant, revoke	Type-check + build; API in v2 §6; browser review 16 Sep

✓	L10 App sidebar layout	✓ Every link opens its page, highlighted, with breadcrumb ✓ Collapse persists across a reload; theme toggle persists	✓ No console errors on any page □ Under-860px drawer (needs a manual look)	Type-check + build; browser review 16 Sep
✓	L11 Design system & FoxByte theme	✓ Tokens for colour, type, space, radius, elevation and motion; components use no raw values ✓ Ember palette in both themes; filled controls legible (<code>--accent-ink</code>) ✓ Fox mark from theme variables, wordmark from brand.json, favicon	✓ Console, Blackbox, Policies and API keys re-laid out; one table style ✓ Every page reviewed in both themes; console checked at 375px □ Not yet reviewed against a running engine with real data	Type-check + build; browser review 21 Sep

M Build, CI & releases 7 of 7 done

ID	Feature · sub-features	Tested by
✓ M1	Makefile targets ✓ build, vet, test, integration suites ✓ Suites run only in a throwaway test VM (guarded)	✓ release, web, WSL builds, feature-doc (both PDFs) CI
✓ M2	Linux CI ✓ Build, vet, test, gofmt ✓ Release script tests, checklist sync check	□ VM integration suites in CI CI
✓ M3	Windows CI ✓ Build, vet, test, Pester, installer encoding ✓ Guard tests read backslash paths, and generated files are pinned to LF (21 Sep)	□ End-to-end WSL install test CI
✓ M4	Releases & GHCR image ✓ Binaries for 5 platforms + fox-verify ✓ SHA256SUMS merged safely between jobs	✓ Fails unless <code>fox update</code> 's files are published ✓ Multi-arch Postgres image on GHCR Script tests in CI
✓ M5	WSL distro & ZFS builds ✓ Prebuilt distro image ✓ Downloaded and verified by the installer (seen on Windows) ✓ Preloaded image tagged as the engine runs it	✓ Imported on a fresh Windows install; preloaded images loaded ✓ Published zstd-compressed, with the gzip size measured beside it in every build log ● zstd unpack on a real Windows install (unit-tested only); the gzip image was 685 MB Actions artifacts; Pester; Windows run 16 Sep
✓ M6	Postgres image ✓ PostgreSQL 18 + wal-g for a fresh install; one image per supported major (16, 18) ✓ The image follows the data: an install on 16 keeps running 16 after an update ✓ A branch on the wrong major is refused before <code>docker run</code>, with the way out	✓ The major lives in one place, and a test holds the Dockerfile, release matrix and Windows preload to it ✓ All three suites pass on PostgreSQL 18.6: 108 + 277 + 51 checks (21 Sep) ✓ A command that moves an install from 16 to 18 (D6), and a portable backup to take first (D5) □ pgvector Unit; all three suites on 18
✓ M7	Product name in one file ✓ brand.json → Go, shell, web and installer blocks, regenerated by <code>make brand</code> ✓ Stored names brand-free and frozen: a rename never migrates an install ✓ Retired env prefixes, state directory, VM names and installs still handled	✓ Guards: retired names anywhere, product names in stored SQL, durable names spelt twice ● The sweep is text substitution: review its diff (it over-reached twice) Unit (brand, gen, oldnames), in CI via go test

N Not started (roadmap)

11 items

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

ID	Item · scope	Related
☐ -	Multi-tenant projects & roles ☐ Projects / organisations ☐ Per-branch permissions (RBAC)	G1, H6
☐ -	Branch merge ☐ Apply one branch's schema changes to another	H18
☐ -	Scheduled backups & retention	D2
☐ -	Automatic HA failover ☐ Health detection, fencing ☐ Multi-host	D4
☐ -	Pipelines: scheduling, incremental / CDC extract, DAG view	J1
☐ -	More import sources: Cassandra, DynamoDB	I1–I8
☐ -	pgvector in the Postgres image	M6
☐ -	Cloud write-once anchors (S3 Object Lock etc.)	H10
☐ -	Release signatures	A9, M4
☐ -	Hosted cloud tier	—
☐ -	Networked storage disaggregation (README roadmap; see the storage engine note on page 1)	C1, D1

Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.